

SMARTROAD

Traffic Accident Risk, Severity & Hotspot Analytics

Computational Analytics - Assignment 2
BCA Semester 7 | Academic Year 2026-2027

Group members: _____

1. Problem, Goal & Data

Problem: recorded traffic collisions contain temporal, road, environmental and geographic signals that are difficult to summarize manually.

Goal: build an end-to-end analytics solution that explores collision patterns, identifies geographic hotspots, and classifies whether a reported collision is high severity.

Dataset: official UK Department for Transport STATS19 collision CSV for 2025. Records processed: 101,525.

2. Analytics Pipeline

Acquire -> Clean -> EDA -> Feature Engineering -> Train/Test -> 5-Fold Cross-Validation -> Evaluation -> Hotspot Clustering -> Dashboard

Python stack: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Streamlit, Plotly.

3. Preprocessing & Feature Engineering

Cleaning: duplicate removal, date/time parsing, numeric validation, median/mode imputation and one-hot encoding.

Engineered features: hour, month, weekend, night, rush_hour, multi_vehicle and modeled road/environment indicators.

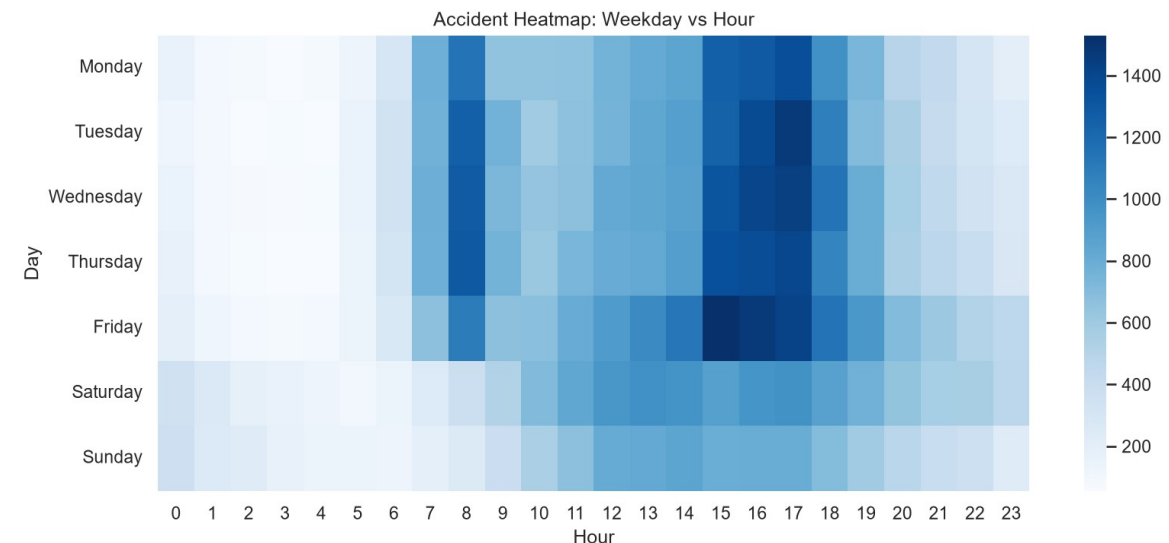
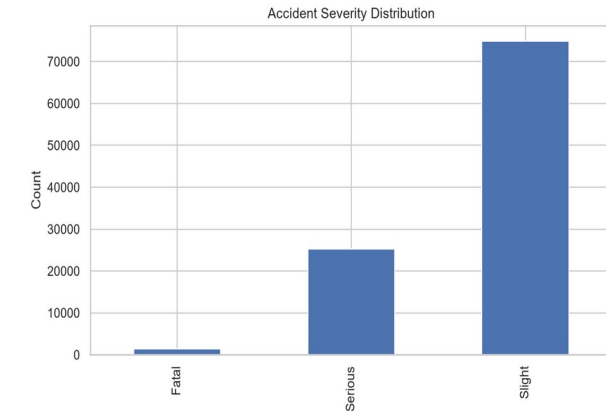
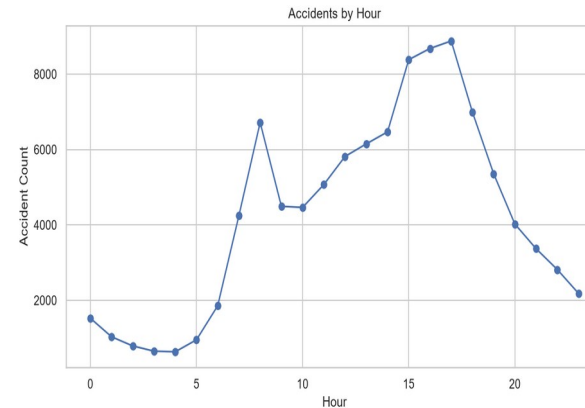
Target: high_severity = Fatal or Serious; 0 = Slight. Target prevalence in the 2025 data: 26.24%.

4. Visual Analytics

Real 2025 visual analytics

- Accident counts by hour reveal strong concentration during daytime and evening periods.
- Severity distribution shows Slight collisions as the largest class, followed by Serious and Fatal.
- The weekday-hour heatmap highlights recurring temporal concentration, especially during weekday peak periods.

Eight real 2025 PNG outputs were generated: six EDA charts plus the confusion matrix and model-comparison chart.



Representative real 2025 EDA outputs

5. Machine Learning Model Comparison

Model	Accuracy	Precision	Recall	F1	CV F1
Random Forest	65.01%	33.85%	34.92%	34.38%	34.59%
Logistic Regression	57.33%	32.60%	58.66%	41.91%	41.85%
Gradient Boosting	60.58%	34.12%	53.95%	41.80%	42.09%

Selected model: Gradient Boosting, based on the highest mean 5-fold cross-validated F1.

6. Final Evaluation

Selected model: Gradient Boosting

Accuracy: 60.58%

Precision: 34.12%

Recall: 53.95%

F1-score: 41.80%

5-fold CV F1: 42.09% $\pm$ 0.24%

The confusion matrix is included to show true/false classifications for the high-severity class. F1 is emphasized because the positive class represents 26.24% of the 2025 records.

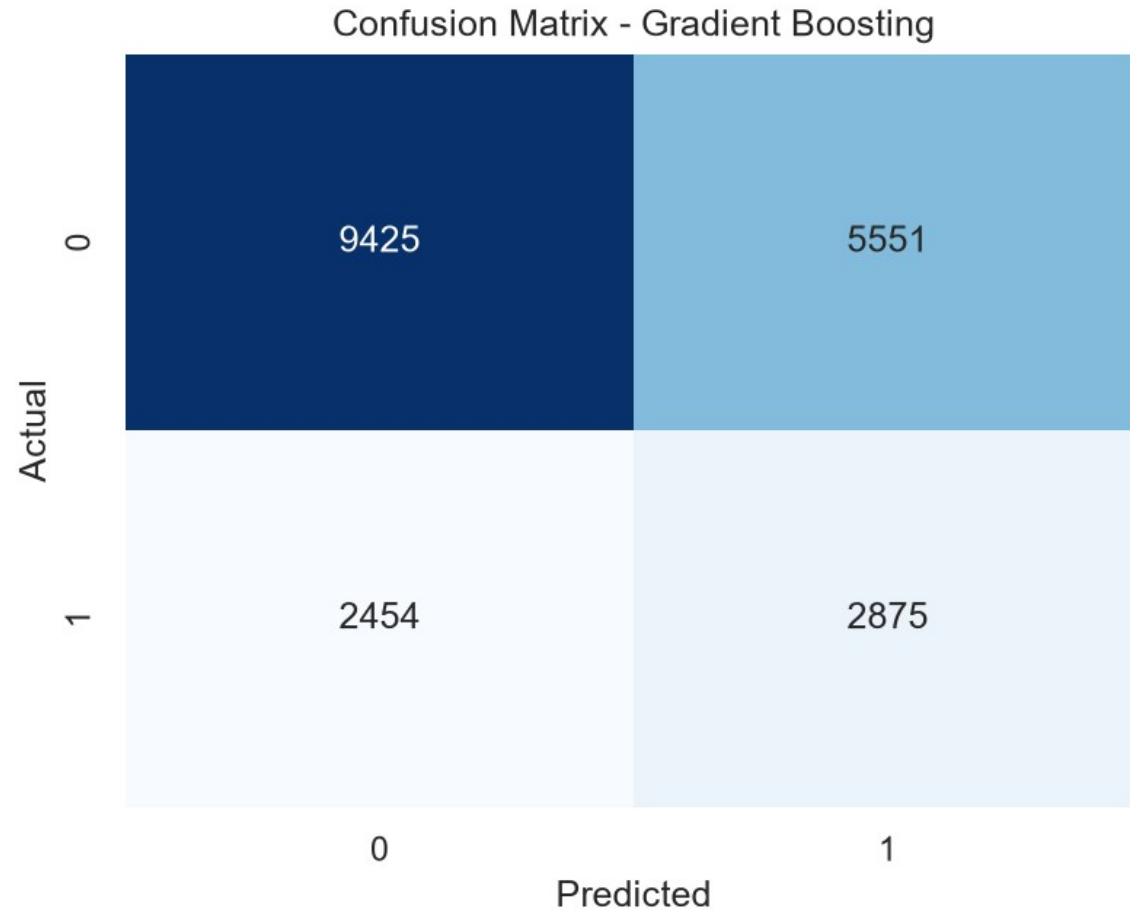


Figure: Gradient Boosting confusion matrix - official 2025 run

7. Feature Importance & Hotspots

Top features in the final 2025 run

1. number_of_vehicles - 0.04165
2. urban_or_rural_area - 0.01993
3. speed_limit - 0.01917
4. road_type - 0.01271
5. hour - 0.00727

K-Means groups valid latitude/longitude records into exploratory geographic clusters and ranks them using collision volume and high-severity rate.

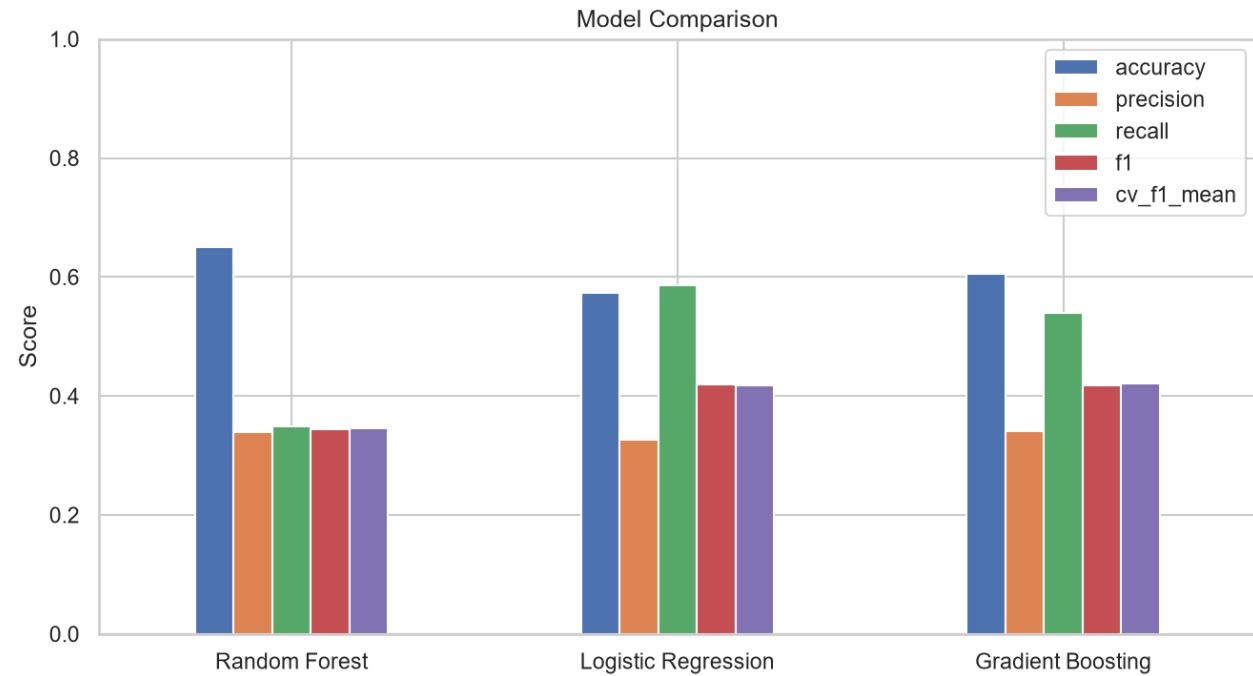


Figure: Three-model comparison - official 2025 run

8. Interactive Dashboard

Streamlit dashboard functions:

- KPI cards for collision count, high-severity count/rate and average vehicles
- Severity and hourly interactive charts
- Geographic visualization
- Adjustable K-Means hotspot clustering
- Model comparison and confusion matrix
- High-severity prediction form
- Data-quality summary

The dashboard uses data/road_collisions_2025.csv when present and otherwise falls back to the demo dataset for local software testing.

9. Conclusion, Deliverables & Future Scope

Conclusion: SmartRoad implements the requested analytics lifecycle from acquisition through predictive modeling, evaluation, clustering and dashboard delivery.

Deliverables: source code, dataset, charts, report, presentation and requirements file.

Future scope: live traffic/weather APIs, exposure-adjusted risk, explainable AI, probability calibration and cloud deployment.

Presentation length: designed for the professor requirement of 8-10 slides and a maximum 10 minutes.